

Lab 2 — Ridge, Lasso and Cross-Validation

Machine Learning I · Tutorial

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Goal of the session. Implement ridge and lasso from the formulas in Lecture 3 — closed form, SVD shrinkage, soft thresholding, coordinate descent — check each against scikit-learn, then choose λ properly and see what regularization does to a logistic regression that has no MLE.

Materials

- `lab_02_student.ipynb` / `lab_02_solution.ipynb`.
- Sources: CS 155 Set 2 (loss functions, effects of regularization, lasso vs. ridge).
- Data: `load_diabetes` (regression) and `load_breast_cancer` (classification), both bundled with scikit-learn.

A note on conventions

Every library defines the penalty slightly differently. Keep this table next to you.

	Objective	Relation to Lecture 3
Lecture 3, ridge	$\ \mathbf{y} - \mathbf{X}\boldsymbol{\beta}\ ^2 + \lambda \ \boldsymbol{\beta}\ ^2$	—
<code>sklearn.linear_model.Ridge(alpha)</code>	$\ \mathbf{y} - \mathbf{X}\mathbf{w}\ ^2 + \alpha \ \mathbf{w}\ ^2$	$\alpha = \lambda$
Lecture 3, lasso	$\frac{1}{2} \ \mathbf{y} - \mathbf{X}\boldsymbol{\beta}\ ^2 + \lambda \ \boldsymbol{\beta}\ _1$	—
<code>sklearn.linear_model.Lasso(alpha)</code>	$\frac{1}{2N} \ \mathbf{y} - \mathbf{X}\mathbf{w}\ ^2 + \alpha \ \mathbf{w}\ _1$	$\alpha = \lambda/N$
<code>LogisticRegression(C)</code>	$\frac{1}{2} \ \mathbf{w}\ ^2 + C \sum_i \log(1 + e^{-\tilde{y}_i \mathbf{x}_i^\top \mathbf{w}})$	$C = 1/\lambda$

Both scikit-learn regressors fit the intercept separately and unpenalized, as Lecture 3 prescribes.

1 Ridge from the Formula (35 min)

Task 1 — closed form

Standardize the diabetes features and centre $\mathbf{y}$. Implement $\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ with `np.linalg.solve`. For $\lambda \in \{0.1, 1, 10, 100\}$, compare with `Ridge(alpha= $\lambda$ , fit_intercept=False)` on the same standardized data. Agreement to 10^{-8} .

Task 2 — the SVD picture

Compute the thin SVD $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ (`np.linalg.svd(X, full_matrices=False)`). Verify numerically that for $\lambda = 10$

$$\hat{\mathbf{y}}_\lambda = \sum_{j=1}^p \mathbf{u}_j \frac{d_j^2}{d_j^2 + \lambda} \mathbf{u}_j^\top \mathbf{y}$$

matches $\mathbf{X}\hat{\boldsymbol{\beta}}_\lambda$ from Task 1. Print the ten shrinkage factors $d_j^2/(d_j^2 + \lambda)$. Which direction is shrunk most, and what is its d_j ? Then plot $\text{df}(\lambda) = \sum_j d_j^2/(d_j^2 + \lambda)$ for $\lambda \in [10^{-2}, 10^4]$ on a log axis. Confirm $\text{df}(0) = 10$.

Task 3 — the ridge path

Plot all ten $\hat{\beta}_j(\lambda)$ against λ on a log axis, $\lambda \in [10^{-2}, 10^4]$. Every curve must go to zero and none may cross zero exactly (ridge never produces an exact zero from a nonzero OLS coefficient — explain why with the SVD formula).

2 Lasso from the Formula (50 min)

Task 4 — soft thresholding

Write `soft(b, lam) = sign(b) (|b| - lam)+`, vectorized. Check the Lecture 3 example: `soft([3, 0.5, -2], 1)` must return `[2, 0, -1]`, and the ridge shrinkage $b/(1 + \lambda)$ must return `[1.5, 0.25, -1]`.

Task 5 — coordinate descent

Implement the lasso by coordinate descent for the Lecture 3 objective $\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1$:

```
def lasso_cd(X, y, lam, iters=20000, tol=1e-10):
    n, p = X.shape
    beta = np.zeros(p)
    r = y.copy()                                     # full residual y - X beta
    for _ in range(iters):
        for j in range(p):
            r_j = r + X[:, j] * beta[j]              # partial residual, coordinate j
                                                         removed
            b_new = soft(X[:, j] @ r_j, lam) / (X[:, j] @ X[:, j])
            r = r_j - X[:, j] * b_new                 # update the residual
                                                         incrementally
            beta[j] = b_new
    return beta
```

Add a stopping rule (sweep until no coordinate moves more than `tol`; correlated columns make plain coordinate descent slow), then compare with `Lasso(alpha=lam/N, fit_intercept=False)` for $\lambda \in \{50, 200, 800\}$ on standardized, centred data. Which coefficients are exactly zero? Verify the KKT condition on the solution: $|\mathbf{x}^{(j)\top} \mathbf{r}| = \lambda$ for every active j and $\leq \lambda$ for every inactive j .

Task 6 — the lasso path and $\lambda_{\max}$

Compute $\lambda_{\max} = \|\mathbf{X}^T \mathbf{y}\|_{\infty}$. Show that `lasso_cd(X, y, 1.01 * lam_max)` returns all zeros and `lasso_cd(X, y, 0.99 * lam_max)` does not. Then plot the full path with `sklearn.linear_model.lasso_path`. In what order do features enter? Compare with the ridge path of Task 3 on the same axes.

3 Choosing λ (35 min)

Task 7 — cross-validation with error bars

For 40 values of λ on a log grid, compute the 10-fold CV MSE of the lasso and its standard error across folds (use `KFold(10, shuffle=True, random_state=155)`; standardize inside each fold with a `Pipeline`). Plot the curve with error bars. Mark $\hat{\lambda}_{\min}$ and the one-standard-error choice $\hat{\lambda}_{1se}$. How many nonzero coefficients does each have?

Task 8 — optimism, measured

Lecture 3 proves $\mathbb{E}[\text{test}] - \mathbb{E}[\text{train}] = 2\sigma^2 \text{df}/N$ for a linear smoother. Test it on ridge: for $\lambda \in \{0.1, 1, 10, 100, 1000\}$, compute (i) training MSE, (ii) 10-fold CV MSE, (iii) $2\hat{\sigma}^2 \text{df}(\lambda)/N$ with $\hat{\sigma}^2$ from the OLS residuals. Tabulate (ii)–(i) against (iii). They should track each other in size and in trend.

4 Regularized Logistic Regression (40 min)

Task 9 — separable data has no MLE

Generate two well-separated Gaussian clouds in $\mathbb{R}^2$ (30 points each, means $(-2, 0)$ and $(2, 0)$, unit variance). Fit `LogisticRegression(penalty=None, max_iter=10000)` and print $\|\hat{\mathbf{w}}\|$ and the fitted probabilities. Then fit with $C \in \{10^3, 10^2, 10, 1, 0.1\}$ and plot $\|\hat{\mathbf{w}}\|$ against C on a log axis. Explain the shape using Lecture 2, slide “When the MLE Does Not Exist”.

Task 10 — a real classifier, evaluated properly

On `load_breast_cancer` ($N = 569$, $p = 30$), with an 80/20 split:

1. Fit ℓ_2 -regularized logistic regression with C chosen by 5-fold CV on the training set.
2. Report accuracy, precision, recall, F_1 and the confusion matrix on the test set at threshold 0.5.
3. Plot the ROC curve and compute the AUC two ways: with `roc_auc_score`, and directly as the fraction of (positive, negative) test pairs with the positive scored higher. They must agree.
4. Suppose a false negative (missed malignancy) costs 10 times a false positive. Compute $\tau^* = c_{FP}/(c_{FP} + c_{FN})$, re-threshold, and report the new confusion matrix and expected cost. Compare with the cost at $\tau = 0.5$.

5 Exercises (to submit before Lab 3)

Exercise 1. Hoerl–Kennard. With $\mathbf{X}^\top \mathbf{X} = \mathbf{V} \mathbf{D}^2 \mathbf{V}^\top$ and $\boldsymbol{\gamma} = \mathbf{V}^\top \boldsymbol{\beta}$, show that the total mean squared error $\mathbb{E} \left\| \hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta} \right\|^2$ of ridge equals $\sum_j [\sigma^2 d_j^2 + \lambda^2 \gamma_j^2] / (d_j^2 + \lambda)^2$, differentiate at $\lambda = 0$, and conclude that its derivative is negative — so some $\lambda > 0$ beats OLS.

Exercise 2. Extend `lasso_cd` to the elastic net $\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \frac{\lambda_2}{2} \|\boldsymbol{\beta}\|^2$ by deriving the coordinate update (it is soft thresholding divided by $\left\| \mathbf{x}^{(j)} \right\|^2 + \lambda_2$). Check against `ElasticNet`.

Exercise 3. Repeat Task 7 with 20 different random fold assignments. Report the mean and standard deviation of $\hat{\lambda}_{\min}$ and of $\hat{\lambda}_{1\text{se}}$ (on the log scale). Which choice is more stable, and why would you expect that from the shape of the CV curve?

Submission

One executed notebook, `lab02_lastname_firstname.ipynb`, to the course drive before Wednesday, September 23, 18:00.